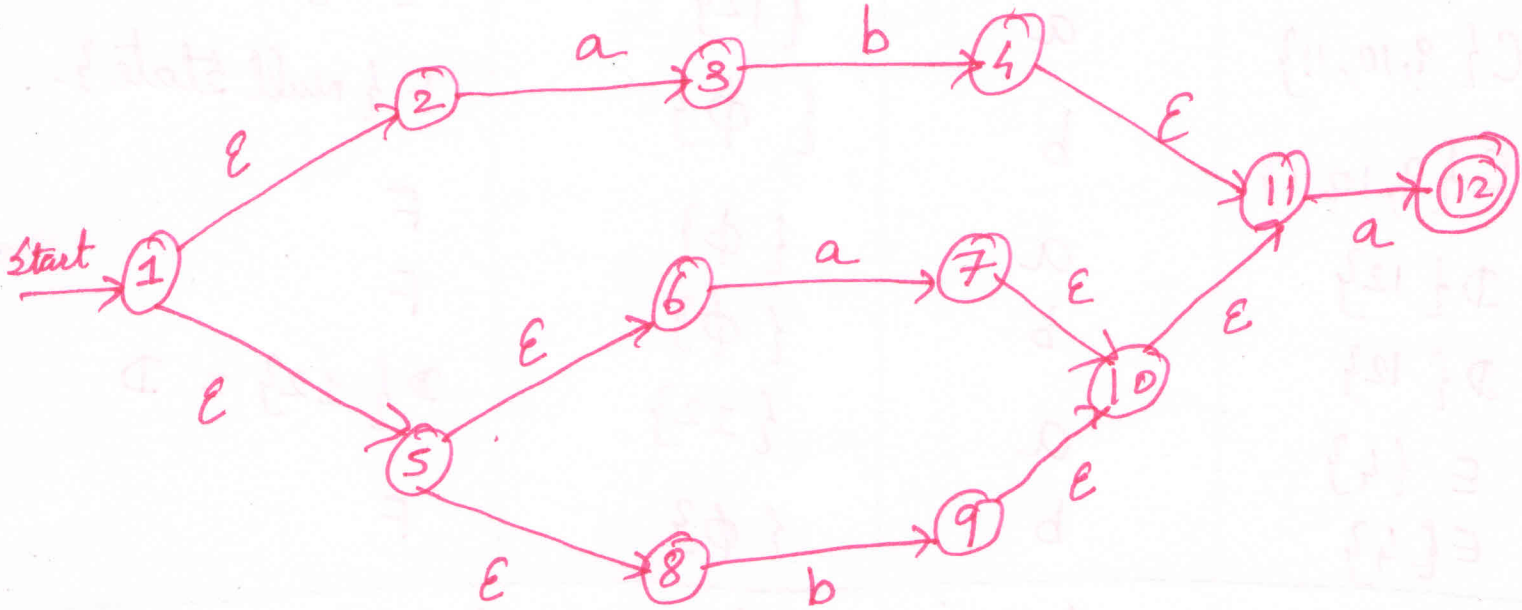
Problem:

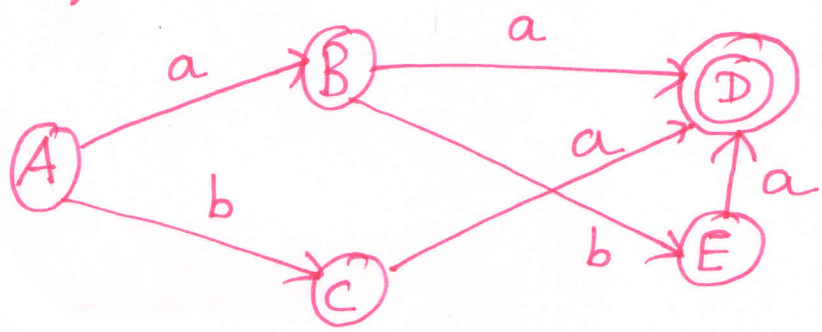
Construct the NFA and DFA that recognizes the regular expression $(ab|(a|b))a$.

Solution:

(i) NFA (Non-Deterministic Finite Automata. for the Regular expression.



(ii) Construction of DFA (Deterministic Finite Automata) for the regular expression is,

Current state	Current I/P	δ(State, Input)	ε-closure(δ(state, I/P)) = new state
A{1}	a	{3,7}	{3,7,10,11} = B
A{1}	b	{9}	{9,10,11} = C
B{3,7,10,11}	a	{12}	{12} = D
B{3,7,10,11}	b	{4}	{4,11} = E
C{9,10,11}	a	{12}	{12} = D
C{9,10,11}	b	{ϕ}	F {null state}.
D{12}	a	{ϕ}	F
D{12}	b	{ϕ}	F
E{4}	a	{12}	D{12} = D
E{4}	b	{ϕ}	F

DFA: for expression (ab|(a|b))a.